

Passive DS-TWR Known-Geometry Calibration

Static receive-only tag validation

UWB Module project

27 July 2026

Abstract

This report validates the project-specific Passive DS-TWR protocol after restoring a surveyed four-anchor geometry. Hardware-bias calibration reduces the live position-bias magnitude from 3.51 cm to 0.96 cm . The embedded solver improves from 4.12 cm to 3.33 cm 2-D RMSE. Re-solving the same calibrated observations with the surveyed geometry reaches 1.89 cm **RMSE** and 3.11 cm **P95** at 49.17 **complete frames/s**. The remaining live accuracy penalty is therefore dominated by autonomous anchor-geometry reconstruction rather than by the passive observation equation.

1 Experimental setup

Four anchors formed a surveyed 3.000 m square and tag 1 was placed at its center:

Node	Coordinate [m]	Node	Coordinate [m]
A2	(0.000, 0.000)	A3	(0.000, 3.000)
A4	(3.000, 0.000)	A5	(3.000, 3.000)
Tag 1	(1.500, 1.500)	Placement tolerance	approximately $\pm 2\text{ mm}$

Both independent captures lasted 120 s and used Robust Rotating scheduling, 5 ms slots, a 1 ms frame gap, 1000 us RESP and FINAL delays, and UWB channel 5. The first capture used raw observations and raw piggybacked DS ranges. The second capture used the calibration derived from the first capture.

2 Position results

Processing path	Positions	Rate [Hz]	Bias X/Y [cm]	RMSE [cm]	P95 [cm]
Uncalibrated, autonomous geometry	3,362	28.00	$+3.24 / -1.35$	4.12	6.09
Calibrated, autonomous geometry	3,512	29.26	$-0.66 / -0.70$	3.33	5.66
Calibrated, surveyed geometry	5,901	49.17	$-0.30 / -0.27$	1.89	3.11

Calibration improves the live embedded RMSE by 19.2% and reduces the systematic position-bias magnitude by 72.7% . The HTTP snapshot collector keeps only the newest embedded position, so its $28\text{--}29\text{ samples/s}$ is a capture limit. Offline reconstruction of one solution per complete three-slot star shows the actual 49.2 frames/s position cadence.

Passive DS-TWR static position cloud

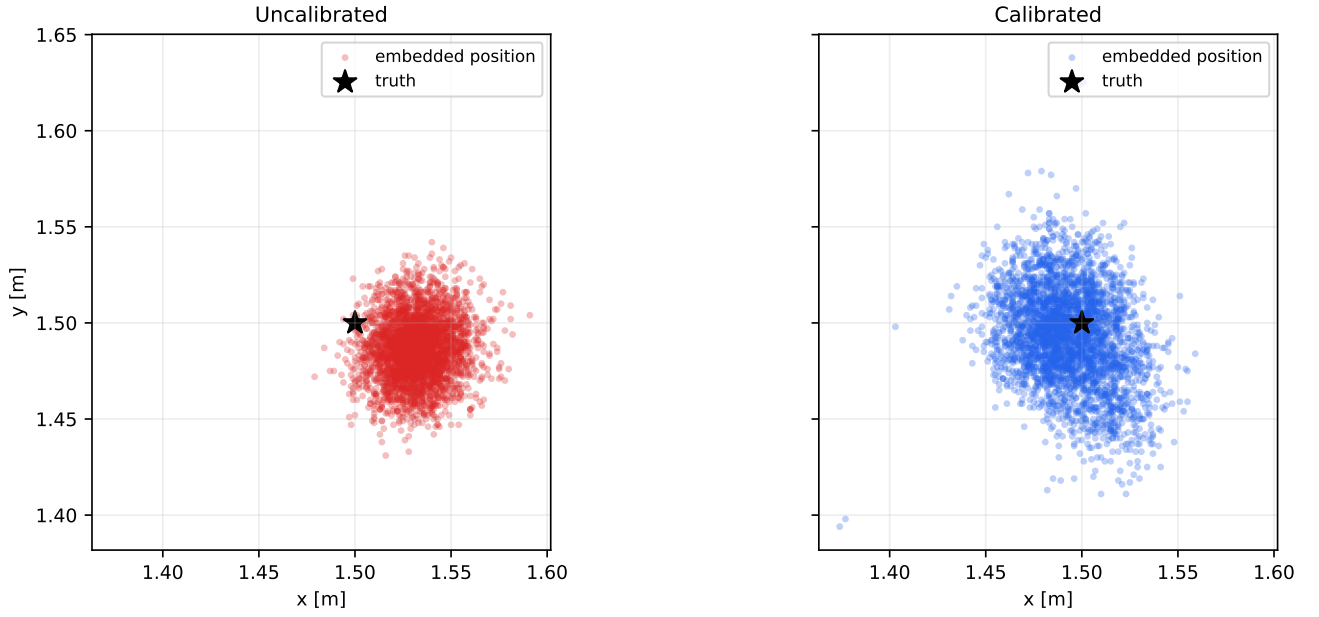


Figure 1: Embedded position clouds before and after calibration. The black star is the surveyed tag position.

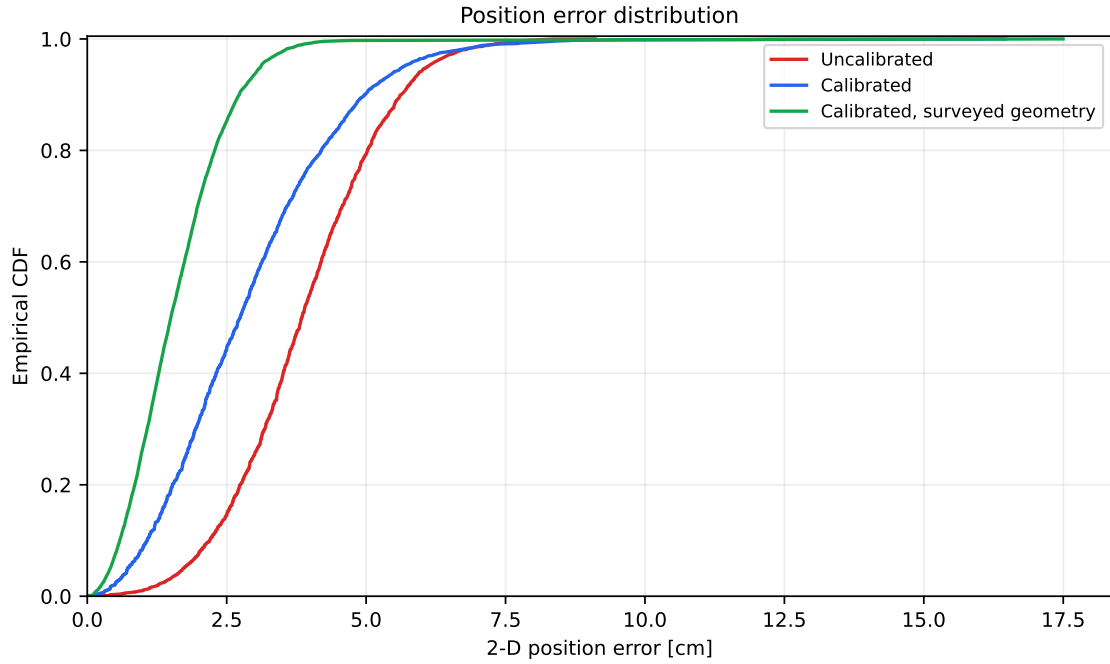


Figure 2: Empirical 2-D error distributions. Replacing only the autonomous geometry with surveyed coordinates exposes the quality of the calibrated passive observations.

3 Calibration model

The scalable directed-observation model is

$$e_{I \rightarrow R} = b_R - b_I,$$

where b_I and b_R are per-anchor biases. A2 fixes the arbitrary common offset. The first half of the uncalibrated observation capture produced:

Anchor	Observation bias	Anchor	Observation bias
A2	0 mm	A3	+34 mm
A4	-16 mm	A5	-55 mm

This uses only $N - 1$ independent values for N anchors while reproducing the corrections for all 12 directed paths. The piggybacked DS ranges use one additional correction per unordered anchor pair:

Pair	Bias	Pair	Bias	Pair	Bias
A2-A3	+71 mm	A2-A4	+56 mm	A2-A5	-49 mm
A3-A4	+67 mm	A3-A5	+66 mm	A4-A5	-29 mm

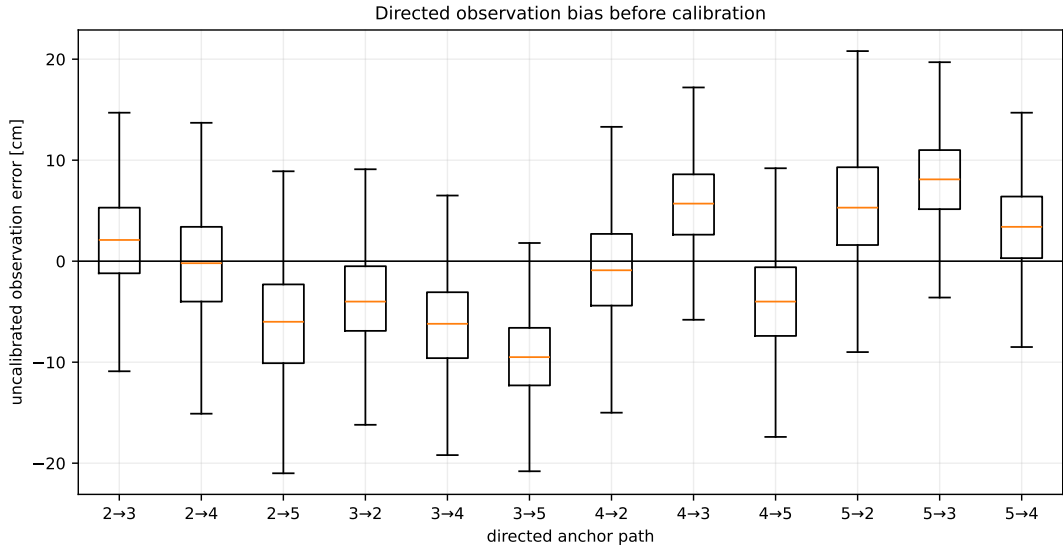


Figure 3: Uncalibrated error on each directed path. Opposite signs are consistent with a compact per-anchor bias model.

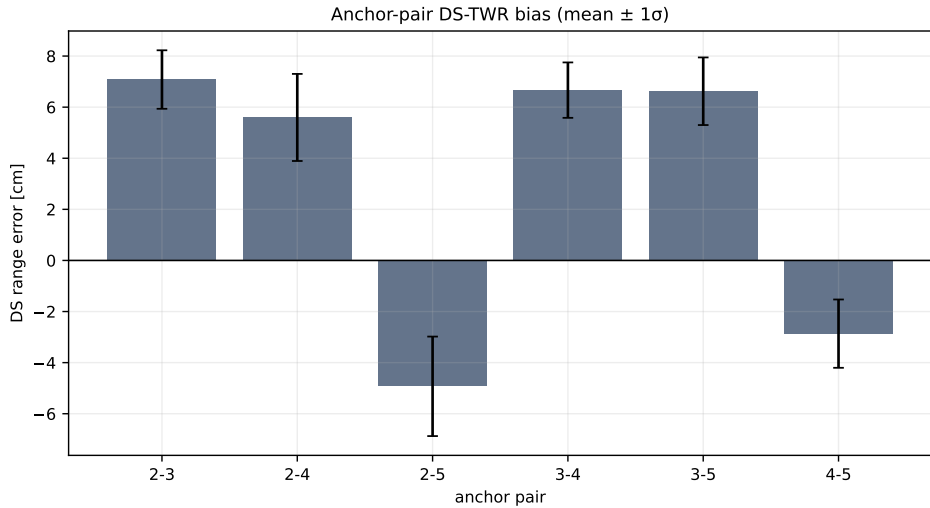


Figure 4: Mean anchor-pair DS range error with one standard deviation.

4 Protocol health and interpretation

During the calibrated capture, anchor radio success ranged from 97.37% to 98.49%. The embedded calibrated distribution contains five estimates above 10 *cm* (0.142%), including two above 15 *cm*. They are short transients and remain included in every reported statistic and plot.

The uncorrected pair distances still fit a planar geometry with 1.99 *cm* range-residual RMS, but the resulting anchor coordinates are displaced by several centimeters. Consequently:

- CFO correction of the responder reply interval is working;
- passive directed observations benefit from the per-anchor bias model;
- piggybacked DS ranges benefit from per-pair corrections;
- autonomous geometry is now the largest remaining live accuracy penalty.

The next implementation step should robustly filter pair ranges before they update geometry: a rolling median or Huber estimate, a physical-change gate, and a slower geometry-update cadence. Validation should then cover several static tag locations and a dynamic trajectory. A single center position identifies the current bias, but cannot establish workspace-wide accuracy.

5 Reproducibility

The repository preserves both compressed JSONL captures, metadata, wireless logs, SHA-256 manifests, CSV metrics, figures, and the deterministic analyzer:

```
python3 tools/uwb_passive_ds_known_geometry_analyze.py
```

The complete machine-readable result is in `analysis_summary.json`; detailed methodology is in `README.md`.